

$\gamma(0) \neq 0$,

(2.3)

convention, n/m is not an

ne that γ is a curve germ
)). If $t^{m+1}b(t)$ has a term
(t^m, t^n). This proves the
□

pe edge. Then there exist
mal matrix A on $\mathbb{R}^3$ such

$$(b_m(0,0) = 0). \quad (2.4)$$

positive coordinate system

$$m(u) + \frac{v^n b_n(u,v)}{n!} \Bigg), \quad (2.5)$$

$a_{3m}(u,v)$.

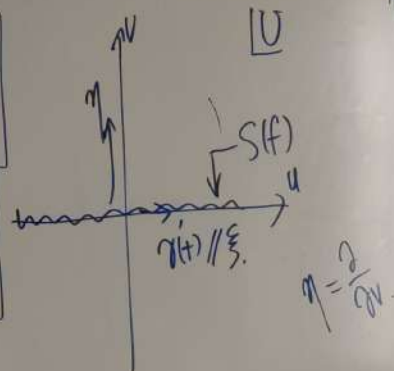
ation on $\mathbb{R}^3$, we may as-
sume $v \mapsto va_{2m}(u,v)^{1/m}$,
 $a_{3m}(u,v)$, $(a_{3m}(0,0) =$
edge, then the function

$$\begin{aligned} y &= x^{\frac{4}{3}} & y &= x^{\frac{m+1}{m}} \\ y' &= \frac{4}{3} x^{\frac{1}{3}} & y' &= \frac{m+1}{m} x^{\frac{1}{m}} \end{aligned}$$

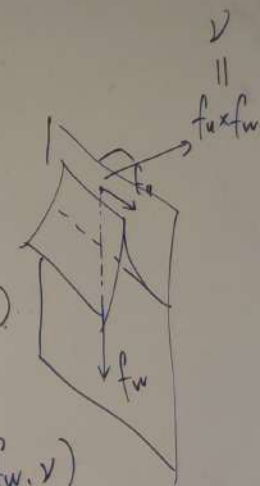
$$K_S = \varepsilon \frac{\det(\hat{\gamma}', \hat{\gamma}'', \nu)}{|\hat{\gamma}'|^3}$$

$$\varepsilon := \text{sgn} \left(\boxed{\det(\gamma', \eta)} \cdot \underbrace{d\lambda(\eta)}_{\eta \lambda} \right)$$

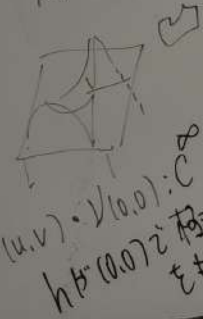
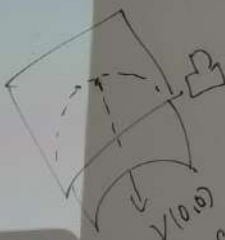
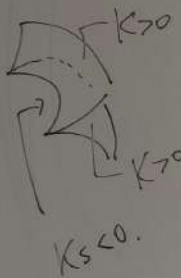
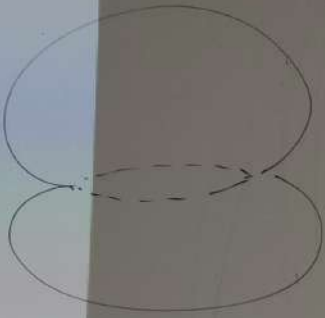
$$\begin{aligned} &(\xi, \eta), (-\xi, -\eta) \\ &\quad : \text{正の向き} \\ &(-\xi, \eta), (\xi, -\eta) \\ &\quad : \text{負の向き} \end{aligned}$$



$$\begin{aligned} \lambda &= \det(f_u, f_v, \nu) \\ \eta \lambda &= \lambda \nu \\ &= \det(f_u, f_v, \nu) \\ &= (f_u \times f_v) \cdot \nu \end{aligned}$$



$$\begin{aligned} m &= 2 \text{ or } 3 \\ K_V &= \frac{\eta'' \cdot \nu}{|\hat{\gamma}'|^2} \\ \Rightarrow \nu &= \frac{\xi \wedge \eta \wedge f}{\| \xi \wedge \eta \wedge f \|} \end{aligned}$$



$K_V, W_{m,m+1} \text{ 12.}$
 $|K_V|, |W_{m,m+1}| : \xi, \eta \text{ 0 1 2 3 4 5 6 7 8 9}$
 $| \Pi_{m,m+1} | : \frac{1}{\| \xi \wedge \eta \|}$
 $| \Pi_{m,m+1} | \text{ 12.}$
 $| \Pi_{m,m+1} | \text{ 12.}$
 $| \Pi_{m,m+1} | \text{ 12.}$

to do: ($\pi_{m,m+i}$ は不変量であることを)

• K の有界性 $\Leftrightarrow \pi_{m,m+1} = \dots = \pi_{m,2m-1} = 0$.

• 凸性の定式化 & 証明.

(Hattoni の 1-2 を再考 (MSSST の定義を正しく))

Lem 27 (改) $j \in \{1, \dots, n-1\}$ とする.

$$\pi_{m,m+1}(u) = \dots = \pi_{m,m+j}(u) = 0 \quad (\forall u)$$

$$\Rightarrow L \tilde{N}_{v^{j+1}}(u, 0) = 0 \text{ である.}$$

$$L \tilde{N}_{v^j}(u, 0) = \boxed{\text{定数} (\neq 0)} \pi_{m,m+j+1}(u).$$

HW • このあたりは Hattoni 解説 1-1 を $m=3$ に限ったものとして整理して 1-1 とする.
 • Lem 27 (改) を $m=3$ に検証する.
 • 凸性の定式化 & 証明.

MSSST

$$\boxed{w_{m,m+i-1} \neq 0}$$

本当は不要.

$\Rightarrow w_{m,m+i}$ は well-def.

M=3 論文

• K とは別に凸性判定量: $\pi_{3,4}(u, v)$ を導入.
 • K の有界性.

Thm D (3.4) - カスミの判定条件.
Cor E 内在的判定条件.

Thm A $\pi_{3,4}$ intrinsic. $\pi_{3,4} = 0 \Rightarrow \pi_{3,5}$ intrinsic.

Thm B K の有界性.

Thm C 凸性.

Appendix

Why $w_{3,4} < w_{3,5}$ は H の有界性.